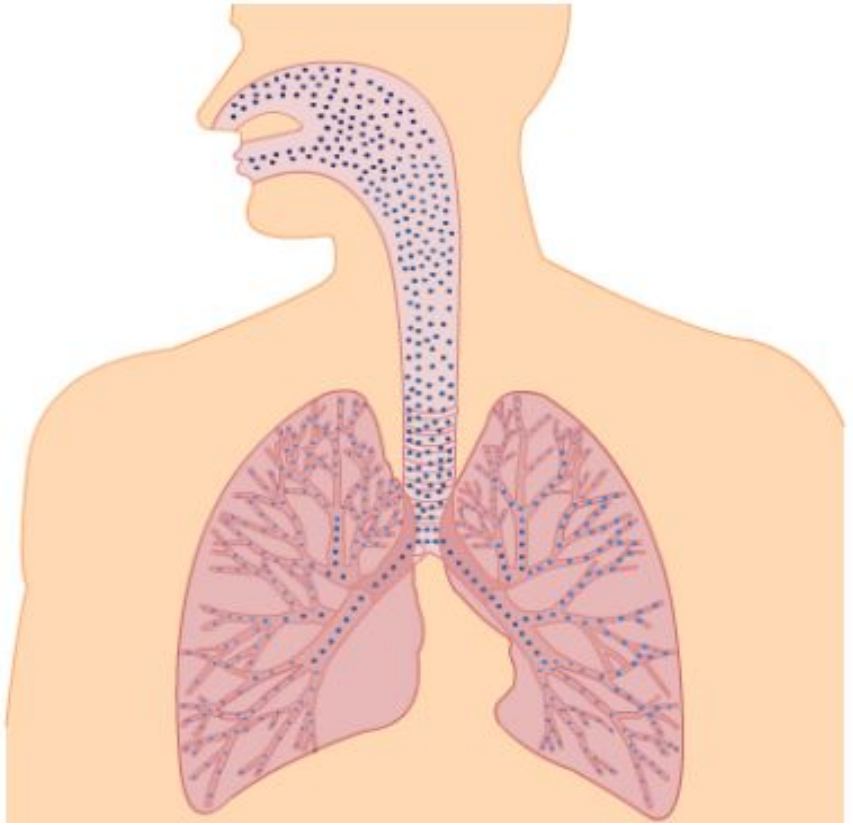
VENTILATION AND ALVEOLAR VENTILATION

DR HUMAIRA FAZAL
LECTURER
DIKIOHS



LEARNING OBJECTIVES:

By the end of lecture students should be able to:

- Define ventilation and alveolar ventilation
- Differences between ventilation and perfusion in different parts of the lungs
- Define Dead space (DS), anatomical dead space, and physiological dead space (total dead space)
- Bohr's equation to estimate total dead space
- Compare the gas compositions and partial pressures of inspired, alveolar and expired air.

VENTILATION:

It refers to circulation of replacement of air or gas in a space.

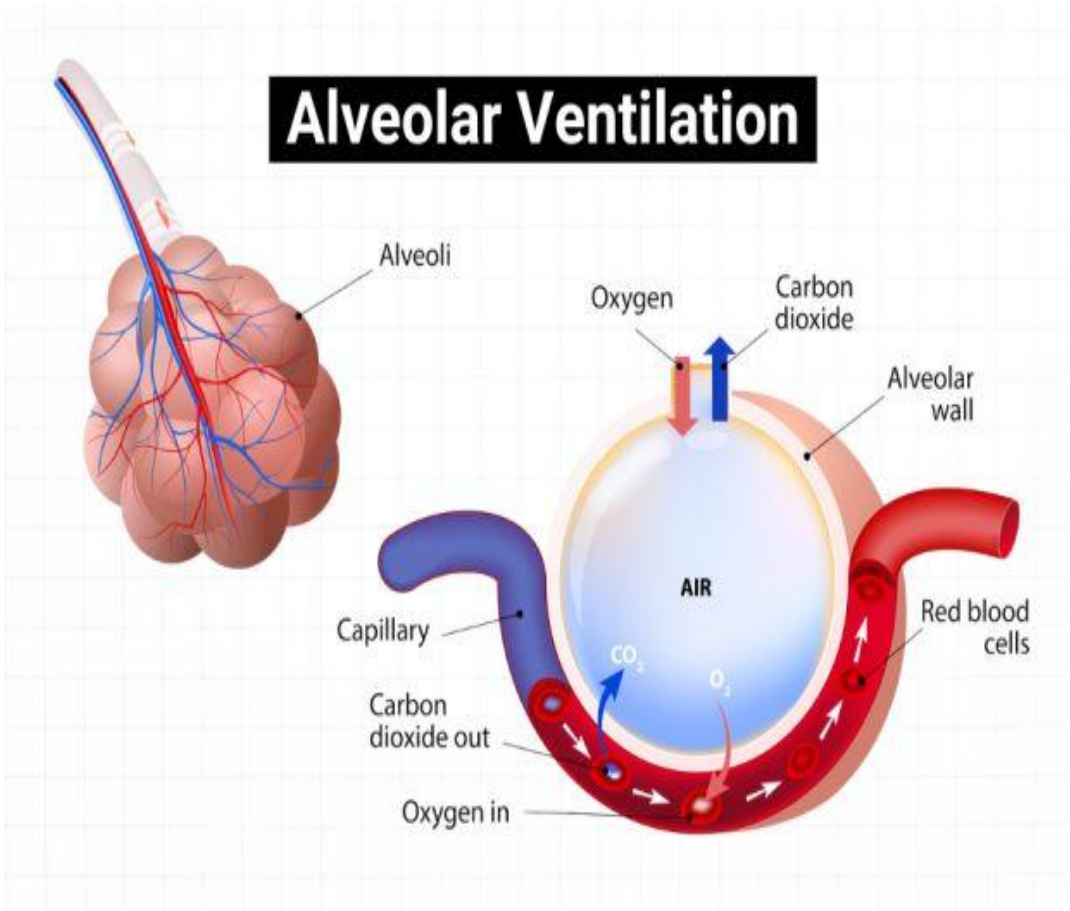
In respiratory physiology, ventilation is the rate at which air enters or leaves the lungs

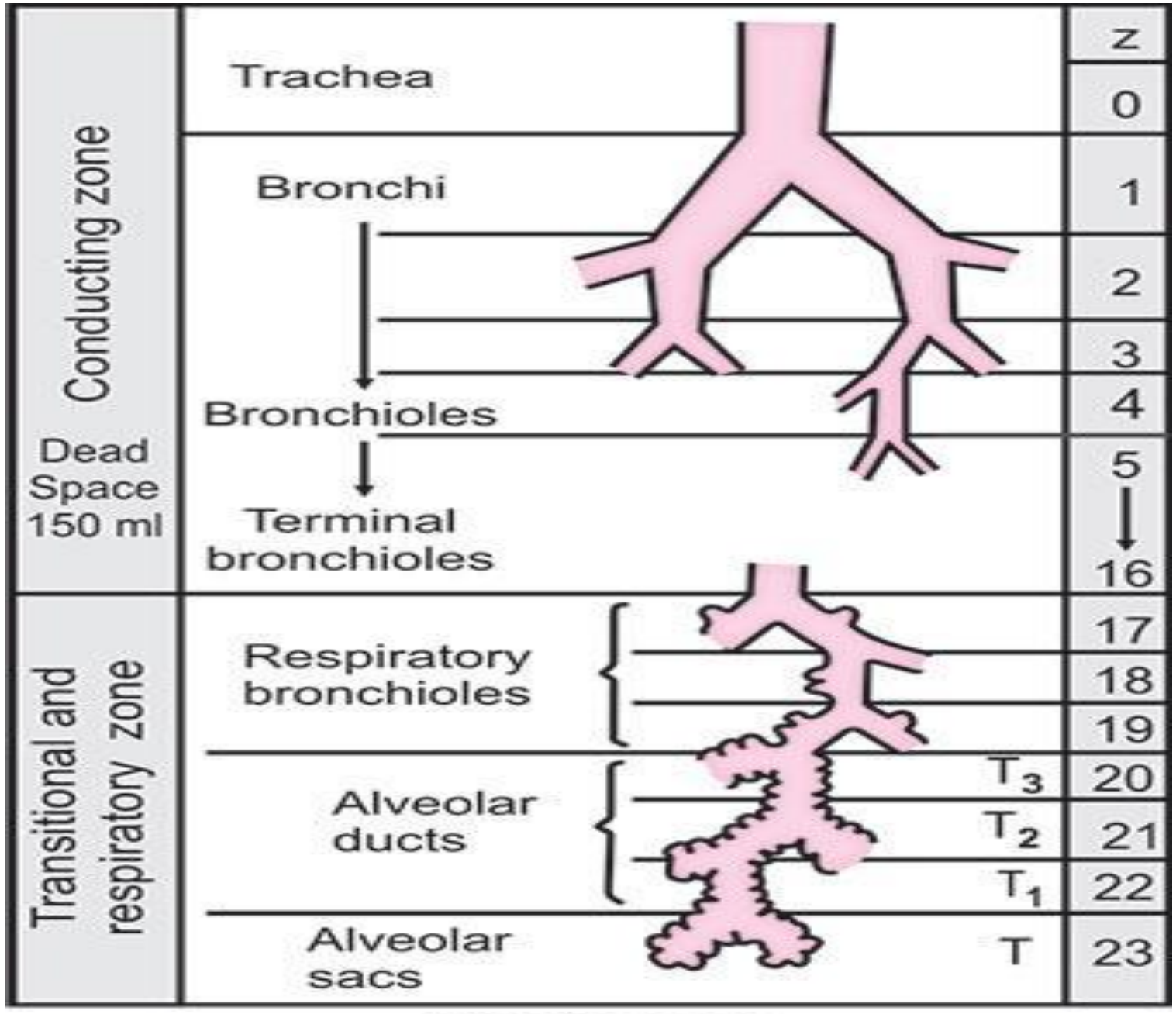
Ventilation in respiratory physiology is of two types:

1. Pulmonary ventilation
2. Alveolar ventilation.

ALVEOLAR VENTILATION:

- Alveolar ventilation is the amount of air utilized for gaseous exchange every minute.
- During inspiration, some of the air inhaled never reaches the gas exchange areas but instead fills the non-gas exchange areas (conducting zone) of the respiratory tract called the dead space, which is equal to about 150mL.





Divisions of
Tracheo-bronchial tree-23 divisions

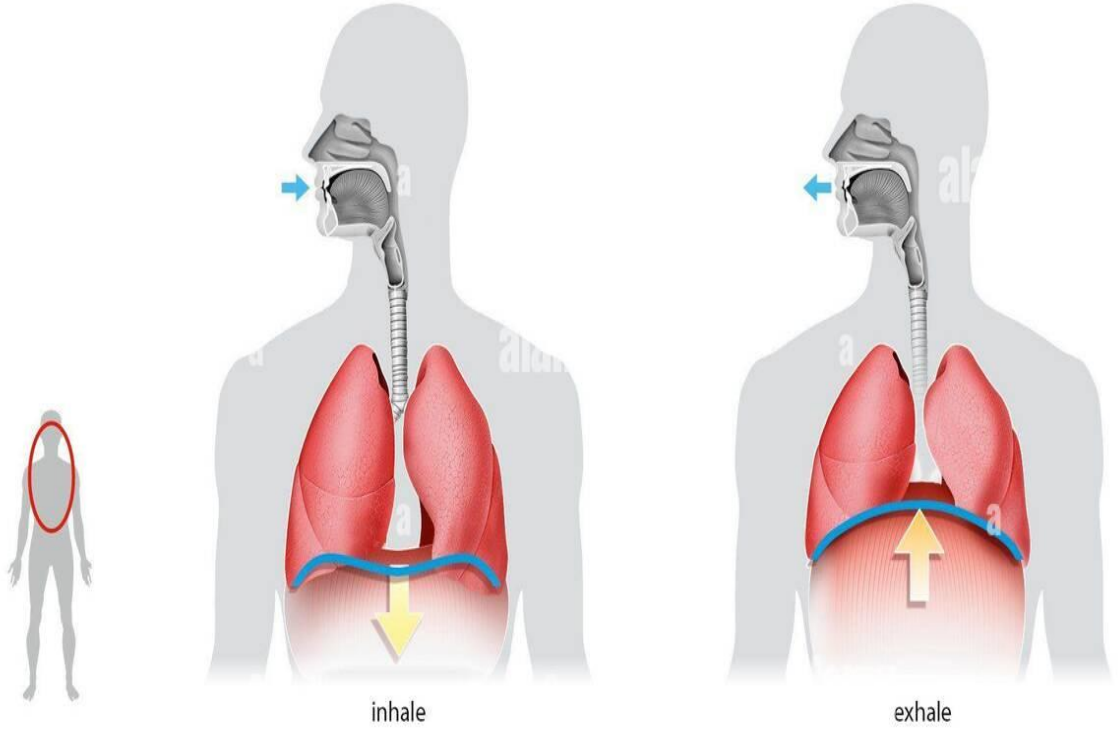
- During expiration, out of 500 mL of tidal volume 150 mL of the alveolar expired air remains in the conducting passages.
- Therefore, 500 mL air entering the lungs only 350 mL/breath is the fresh air which contributes to the alveolar ventilation.

NORMAL VALUE AND CALCULATION:

- Normal value of alveolar ventilation is 4,200 mL (4.2 L)/ minute.
- It is calculated by the formula:
- Alveolar ventilation = (Tidal volume – Dead space) x Respiratory rate = (500 – 150) mL × 12/minute = 4,200 mL (4.2 L)/minute.

PULMONARY VENTILATION:

- Pulmonary ventilation is defined as the volume of air moving in and out of respiratory tract in a given unit of time during quiet breathing. It is also called minute ventilation or respiratory minute volume (RMV).
- Pulmonary ventilation is a cyclic process, by which fresh air enters the lungs and an equal volume of air leaves the lungs.



NORMAL VALUE AND CALCULATION:

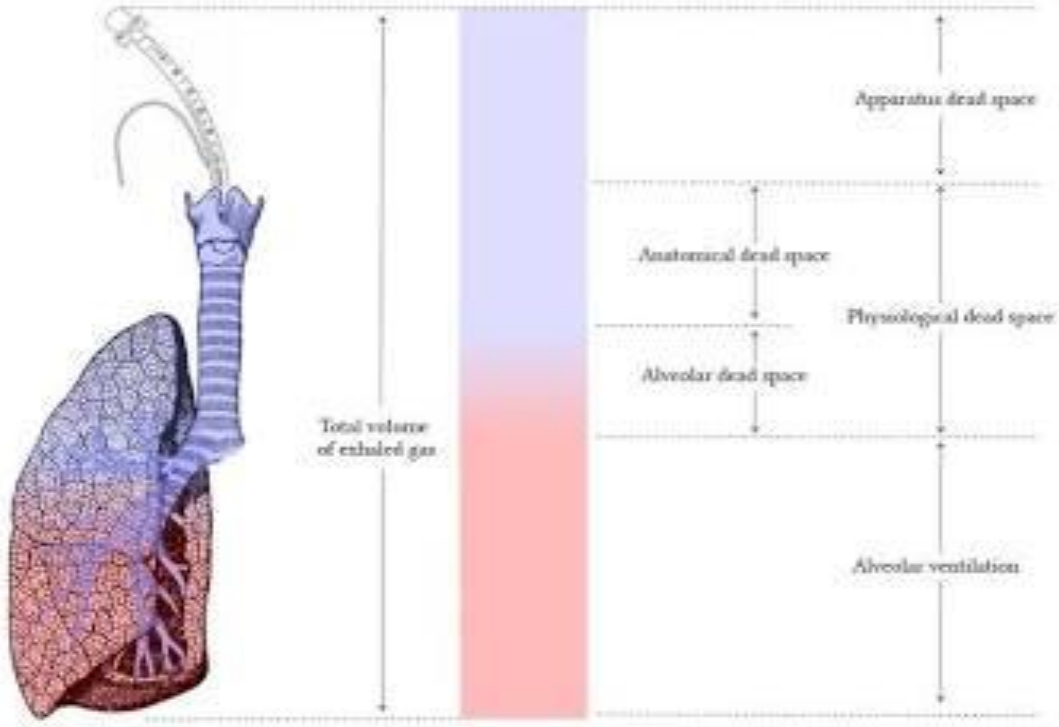
- Normal value of pulmonary ventilation is 6,000 mL (6 L)/minute.
- It is the product of tidal volume (TV) and the rate of respiration (RR). It is calculated by the formula: Pulmonary ventilation = Tidal volume × Respiratory rate = 500 mL × 12/minute = 6,000 mL/minute

DEAD SPACE:

- Dead space is defined as the part of the respiratory tract, where gaseous exchange does not take place. Air present in the dead space is called dead space air.
- TYPES OF DEAD SPACE:

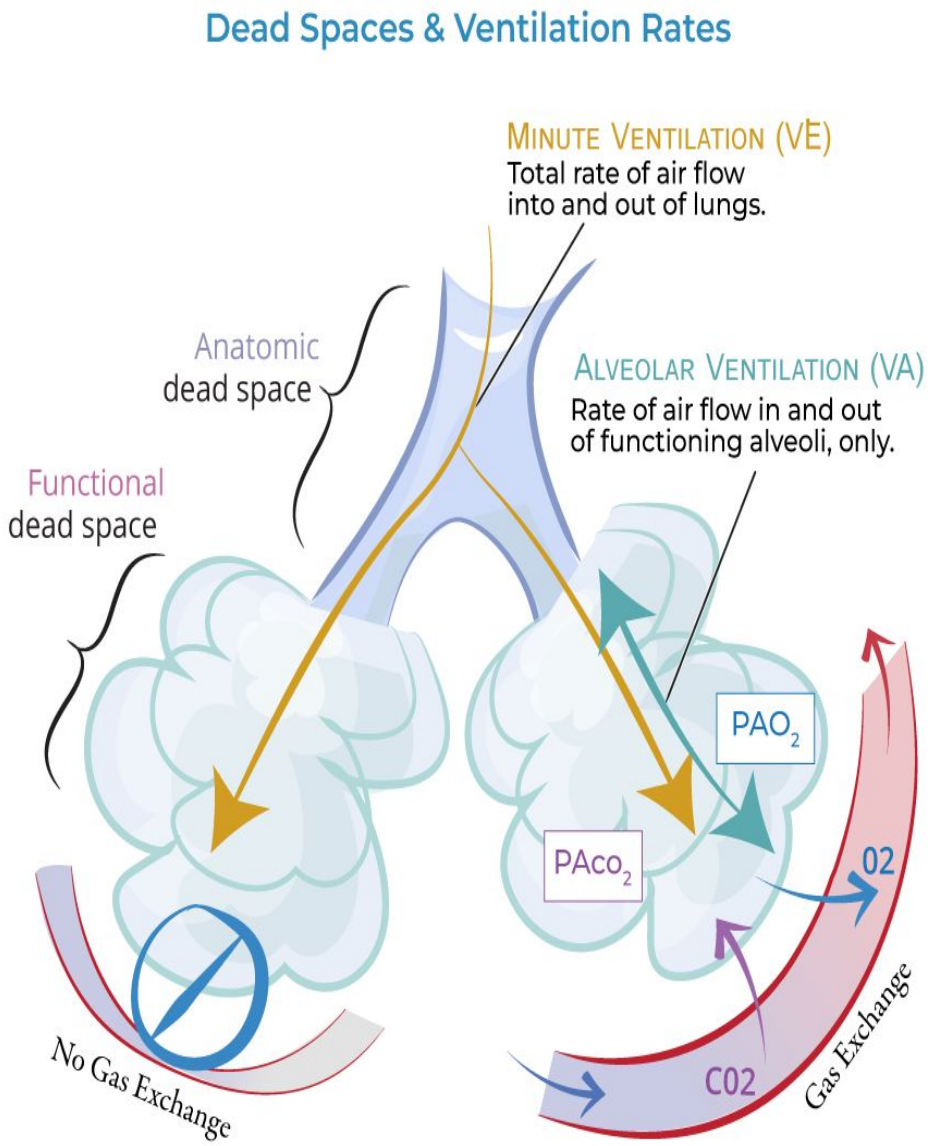
Dead space is of two types:

1. Anatomical dead space
2. Physiological dead space



ANATOMICAL DEAD SPACE:

- Anatomical dead space extends from nose up to terminal bronchiole. It includes nose, pharynx, trachea, bronchi and branches of bronchi up to terminal bronchioles.
- These structures serve only as the passage for air movement. Gaseous exchange does not take place in these structures.



ALVEOLAR DEAD SPACE:

- It refers to the volume of air present in those alveoli which do not take part in gas exchange.
- Normally, all the alveoli take part in the gas exchange, but in some lung diseases, some alveoli do not take part in the gas exchange.

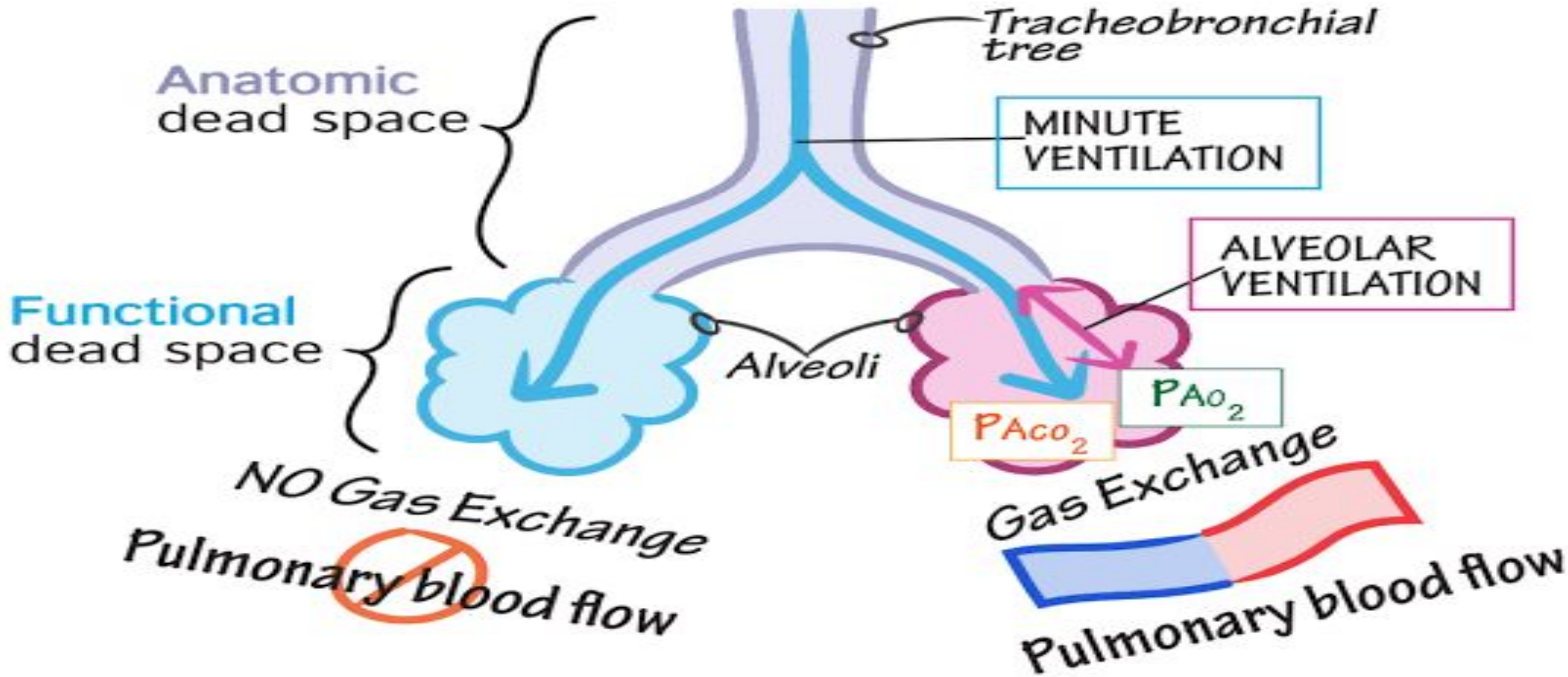
PHYSIOLOGICAL DEAD SPACE:

- Physiological dead space includes anatomical dead space plus two additional volumes.
- Additional volumes included in physiological dead space are:

1. Air in the alveoli, which are non-functioning. In some respiratory diseases, alveoli do not function because of dysfunction or destruction of alveolar membrane.

2. Air in the alveoli, which do not receive adequate blood flow. Gaseous exchange does not take place during inadequate blood supply. These two additional volumes are generally considered as wasted ventilation.

Physiologic Dead Space



Measurement of Total Dead Space:

- Total dead space is calculated using **Bohr's equation**, which relies on the partial pressure of CO2 in expired air (PeCO2)the partial pressure of CO2 in arterial blood PaCO2 or alveolar PCO2, and the tidal volume.

$$\text{Dead space air} = \text{Expired air} \times \frac{\text{Alveolar } CO_2\% - \text{Expired air } CO_2\%}{\text{Alveolar } CO_2\%}$$

- Expired air: 500 ml
  PNG
- Expired air CO_2: 4.0 ml%
  PNG
- Alveolar CO_2: 5.5 ml%
  PNG

$$\text{Dead space air} = 550 \times \frac{(5.5 - 4.0)}{5.5} = 150 \text{ ml}$$

Wasted ventilation and wasted air

- Wasted ventilation is the volume of air that ventilates physiological dead space.
- Wasted air refers to air that is not utilized for gaseous exchange. Dead space air is generally considered as wasted air.

NORMAL VALUE OF DEAD SPACE:

- Volume of normal dead space is 150 ml. Under normal conditions, physiological dead space is equal to anatomical dead space. It is because, all the alveoli are functioning and all the alveoli receive adequate blood flow in normal conditions. Physiological dead space increases during respiratory diseases, which affect the pulmonary blood flow or the alveoli.

WASTED AIR AND WASTED BLOOD;

- Ventilation perfusion ratio is not perfect because of existence of two factors on either side of alveolar membrane.
- These factors are:

1. Physiological dead space, which includes wasted air
2. Physiological shunt, which includes wasted blood

- Both wasted blood and wasted air exist on either side of alveolar membrane and both affect the ventilation-perfusion ratio.

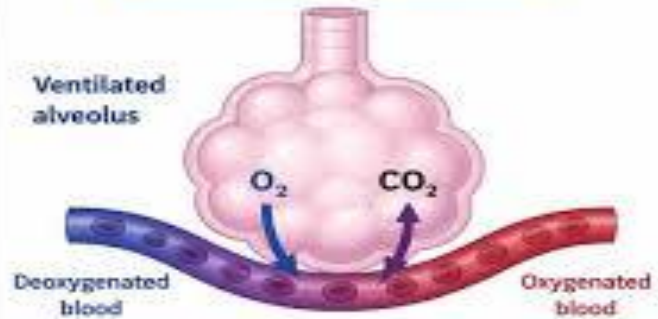
PHYSIOLOGIC SHUNT:

- **It is defined as:**
  "It is a diversion through which the venous blood is mixed with arterial blood."
- Whenever VA/Q is below normal, there is inadequate ventilation to provide the oxygen needed to fully oxygenate the blood flowing through the alveolar capillaries.
- The fraction of venous blood passing through pulmonary capillaries which doesn't get oxygenated is called **"shunted blood"**.
- The total quantitative amount of shunted blood per minute is called the **physiologic shunt**.

RESPIRATORY SHUNT

Blood that passes from the right side to the left side of the circulation without being oxygenated in the lungs.

NORMAL GAS EXCHANGE



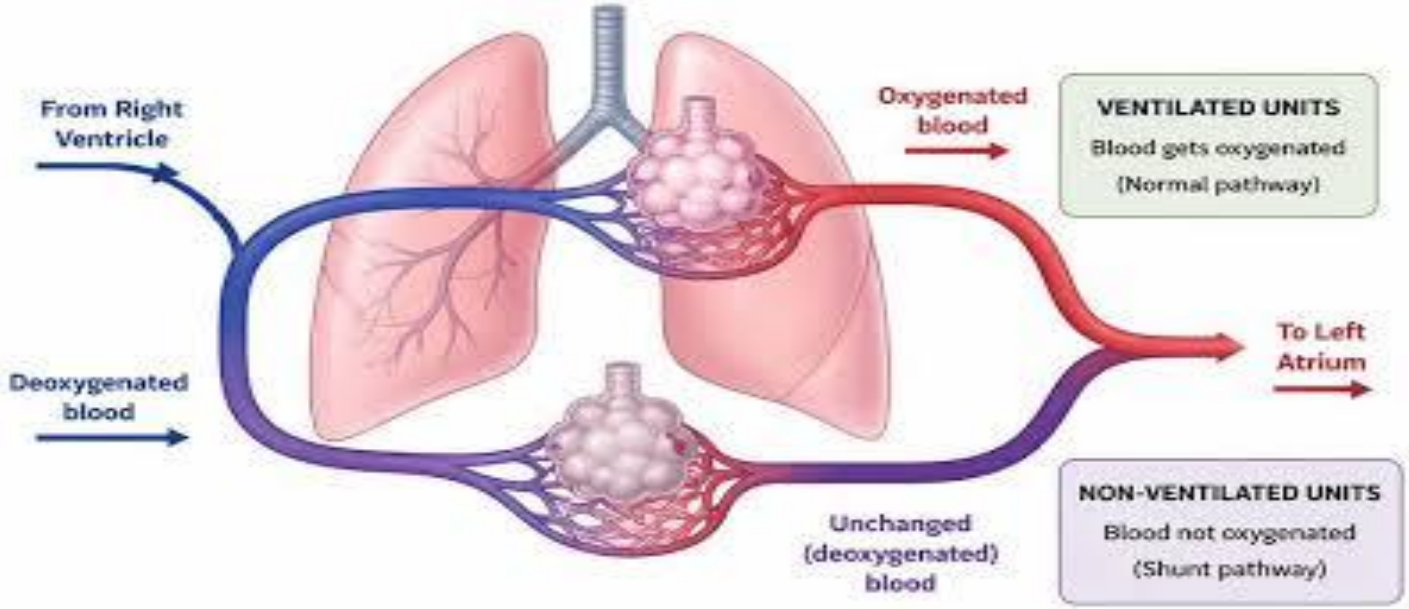
Ventilated alveoli transfer O_2 to blood and remove CO_2.

ALVEOLAR COLLAPSE / FLUID / CONSOLIDATION



No ventilation → No gas exchange.
Blood passes through unchanged.

PULMONARY CIRCULATION



VENTILATED UNITS
Blood gets oxygenated
(Normal pathway)

NON-VENTILATED UNITS
Blood not oxygenated
(Shunt pathway)

RESPIRATORY SHUNT =
Blood flow to non-ventilated areas of the lung that drains into the left heart without being oxygenated.

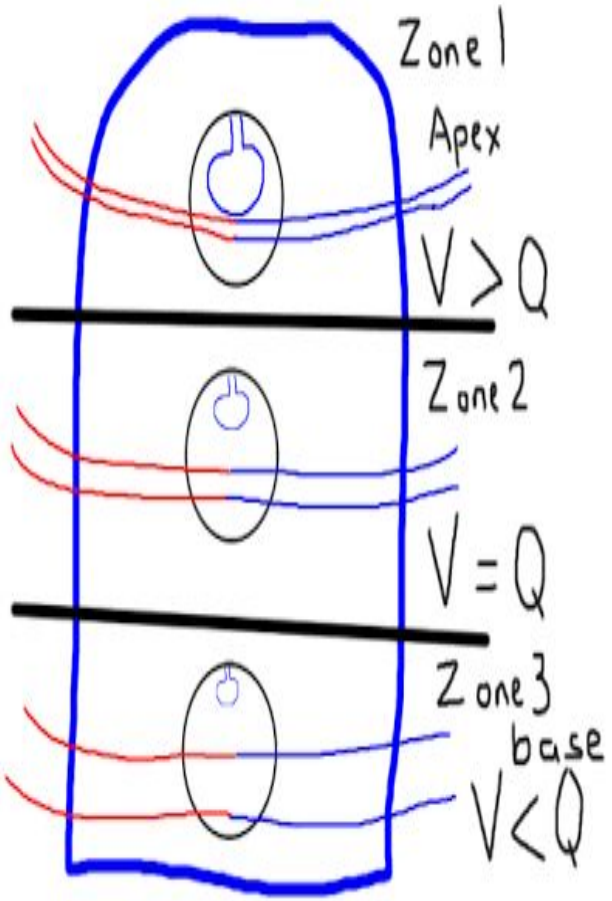
CAUSES OF INCREASED SHUNT

- Atelectasis
- Pneumonia
- Pulmonary edema
- ARDS
- Consolidation

Shunt reduces overall oxygenation because deoxygenated blood mixes with oxygenated blood, lowering arterial oxygen content.

VENTILATION-PERFUSION RATIO:

- DEFINITION: Ventilation perfusion ratio is the ratio of alveolar ventilation and the amount of blood that perfuse the alveoli. It is expressed as VA/Q. VA is alveolar ventilation and Q is the blood flow (perfusion).
- Normally, alveolar ventilation (VA) is 4.2-5.0 and the pulmonary blood flow (equal to cardiac output) is approximately 5l/min.
- So, the normal VA/Q is about 0.84 - 0.9. At this ratio maximum oxygenation occurs.



Calculation

Alveolar ventilation is calculated by the formula:

$$\text{Ventilation-perfusion ratio} = \frac{\text{Alveolar ventilation}}{\text{Pulmonary blood flow}}$$

$$\text{Alveolar ventilation} = (\text{Tidal volume} - \text{Dead space}) \times \text{Respiratory rate}$$

$$= (500 - 150)\ \text{mL} \times 12/\text{minute}$$

$$= 4{,}200\ \text{mL/minute}$$

Blood flow through alveoli

$$\text{(Pulmonary blood flow)} = 5{,}000\ \text{mL/minute}$$

Therefore,

$$\text{Ventilation-perfusion ratio} = \frac{4{,}200}{5{,}000}$$

$$= 0.84$$

EFFECTS OF GRAVITY ON THE VENTILATION–PERFUSION (VA/Q) RATIO:

- **Gravitational Influence:** Due to the effects of gravity, the basal alveoli (at the base of the lungs) receive more blood flow and are **overperfused**, while the apical alveoli (at the apex) are **underperfused**.
- **Linear Gradient in Blood Flow:** There is a nearly linear decrease in pulmonary blood flow as you move from the base of the lungs up to the apex.

- **Linear Decrease in Ventilation:** Alveolar ventilation also decreases linearly from the base to the apex. As a result, the basal alveoli are **over ventilated**, while the apical alveoli are **underventilated**.
- **Perfusion vs. Ventilation Impact:** Gravity affects pulmonary blood flow (perfusion) much more significantly than it affects ventilation. Consequently, the apical alveoli are *more underperfused than they are underventilated*.
- **Resulting Ratio:** Because of this relationship, the overall ventilation–perfusion (VA/Q) ratio is **greater than one** at the apex of the lung.

EFFECTS OF ALTERATIONS IN THE (VA/Q) RATIO:

- **Normal VA/Q Ratio:** A normal ventilation–perfusion ratio implies that both alveolar ventilation and pulmonary perfusion are optimal.
- **Optimal Gas Exchange:** Under these normal conditions, gas exchange is highly efficient, maintaining:
- **Alveolar pO2:** approximately 104 mmHg
- **Alveolar pCO2:** approximately 40 mmHg

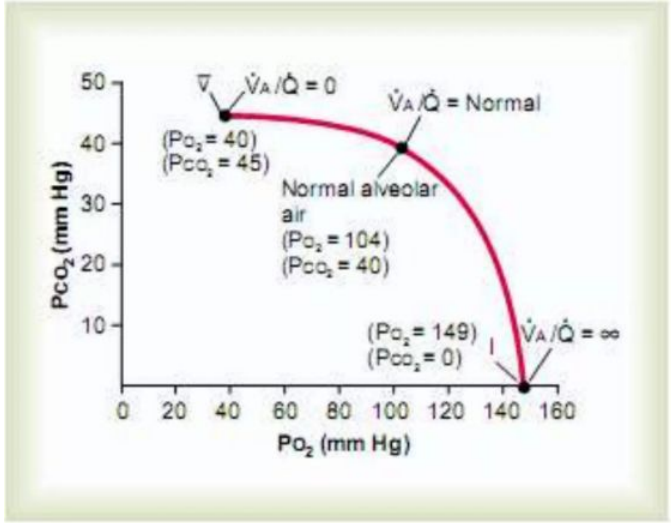


Figure 39–11

Normal P_{O_2}-P_{CO_2}, $\dot{V}_A/\dot{Q}$ diagram.

WHEN VENTILATION EXCEEDS PERFUSION (VA/Q)>NORMAL:

- More air reaches the alveoli than blood flow can process, so air is wasted.
- RESULT: Creates alveolar dead space air and changes alveolar air composition.

EXTREME CASE : ZERO BLOOD FLOW (VA/Q= INFINITY)

- **No Gas Exchange**: Because blood flow drops to zero, no gas exchange occurs.
- **Air Composition**: Alveolar air becomes identical to **humidified inspired air**:
- **Alveolar Po2 = 149mmHg**
- **Alveolar pCO2=** OmmHg

WHEN PERFUSION EXCEEDS VENTILATION(VA/Q< NORMAL):

- Blood flow is greater than airflow.
- Shunted Blood: Blood passes through capillaries without getting enough oxygen.
- Physiological Shunt: Combination of this unoxygenated blood and bronchial vein blood approx 2% of cardiac output,
- Impact: A larger physiological shunt means more blood leaves the lungs without oxygenation.

SIGNIFICANCE OF VENTILATION PERFUSION RATIO:

- **Ventilation-Perfusion Ratio & Gaseous Exchange**: The ventilation-perfusion ratio represents the efficiency of gaseous exchange, depending on a balance between air movement in the lungs and pulmonary blood flow.
- **Factors Affecting the Ratio**: This ratio changes whenever there is an alteration in either **alveolar ventilation** (the amount of fresh air reaching the alveoli) or **blood flow** (perfusion through the capillaries).
- **Ventilation Without Perfusion (Dead Space)**: This describes the scenario where air reaches the alveoli, but there is no blood flow through the surrounding capillaries to exchange gases, rendering that ventilation useless.
- **Perfusion Without Ventilation (Shunt)**: This describes the opposite scenario where blood flows normally through the capillaries, but the corresponding alveoli receive no fresh air, causing blood to bypass gas exchange entirely.

VARIATIONS IN VENTILATION- PERFUSION RATIO:

Physiological Variation

1. Ratio increases, if ventilation increases without any change in blood flow

2. Ratio decreases, if blood flow increases without any change in ventilation

3. In sitting position, there is reduction in blood flow in the upper part of the lungs (zone 1) than in the lower part (zone 3). Therefore, in zone 1 of lungs ventilation perfusion ratio increases three times. At the same time, in zone 3 of the lungs, because of increased blood flow ventilation-perfusion ratio decreases

Pathological Variation:

- In chronic obstructive pulmonary diseases (COPD), ventilation is affected because of obstruction and destruction of alveolar membrane. So, ventilation perfusion ratio reduces greatly.
- In emphysema and chronic smokers, there is low VA/Q ratio, there is inadequate ventilation but there is perfusion.
- When there is inadequate ventilation, there will be maximum attempt for exchange of gases i.e. till the PO_2 and PCO_2 in the alveolar air becomes near to the values in the pulmonary capillary blood.
- So there will be shunted blood which means some of the blood goes without exchange and PO_2 and PCO_2 in the alveolar air resembles the values in the venous blood

- INSPIRED AIR: Inspired air is the atmospheric air, which is inhaled during inspiration.
- ALVEOLAR AIR: Alveolar air is the air present in alveoli of lungs. It is equivalent to the tidal volume minus the dead space. i.e: (500-150) or 350mL.
- Alveolar Air Vs Inspired: Air Alveolar air is different from inspired air in four ways:

1. Alveolar air is partially replaced by the atmospheric air during each breath
2. Oxygen diffuses from the alveolar air into pulmonary capillaries constantly
3. Carbon dioxide diffuses from pulmonary blood into alveolar air constantly
4. Dry atmospheric air is humidified, while passing through respiratory passage before entering the alveoli

TABLE 123.1: Composition of inspired air, alveolar air and expired air

Air	Inspired (atmospheric) air		Alveolar air		Expired air	
Gas	Content (mL%)	Partial pressure (mm Hg)	Content (mL%)	Partial pressure (mm Hg)	Content (mL%)	Partial pressure (mm Hg)
Oxygen	20.84	159.00	13.60	104.00	15.70	120.00
Carbon dioxide	0.04	0.30	5.30	40.00	3.60	27.00
Nitrogen	78.62	596.90	74.90	569.00	74.50	566.00
Water vapor, etc.	0.50	3.80	6.20	47.00	6.20	47.00
Total	100.00	760.00	100.00	760.00	100.00	760.00

RENEWAL:

- Alveolar air is constantly renewed. Rate of renewal is slow during normal breathing. During each breath, out of 500 mL of tidal volume only 350 mL of air enters the alveoli and the remaining quantity of 150 mL (30%) becomes dead space air. Hence, the amount of alveolar air replaced by new atmospheric air with each breath is only about 70% of the total alveolar air.

  Thus,

  Alveolar air = 350 × 100 divided by 500 = 70%

- Slow renewal of alveolar air is responsible for prevention of sudden changes in concentration of gases in the blood.

EXPIRED AIR:

- DEFINITION: Expired air is the amount of air that is exhaled during expiration. It is a combination of dead space air and alveolar air.
- COMPOSITION: Concentration of gases in expired air is somewhere between inspired air and alveolar air.
- METHOD OF COLLECTION: Expired air is collected by using Douglas bag.

THANK YOU